**2.Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with o and end with 1**

# Aim

To write a C program to simulate a Non-Deterministic Finite Automaton (NFA) that accepts strings over the alphabet $\{0, 1\}$ which start with `0` and end with `1`.

# Procedure

1. Start the program.
2. Read the input string containing `0` and `1`.
3. Initialize the NFA with the initial state `q0`.
4. Read each input symbol and update the set of possible current states according to the NFA transitions.
5. Continue until all input symbols are processed.
6. Check whether the final state `q2` is among the possible current states.
7. If `q2` is present, print String Accepted; otherwise, print String Rejected.
8. Stop the program.

## Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i;
    int q0, q1, q2;

    printf("Enter a binary string: ");
    scanf("%99s", str);

    q0 = 1;
    q1 = 0;
    q2 = 0;

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] != '0' && str[i] != '1') {
            printf("Invalid string\n");
            return 0;
```

```
        }

        int n0 = 0, n1 = 0, n2 = 0;

        if (q0 && str[i] == '0') {
            n1 = 1;
        }

        if (q1) {
            if (str[i] == '0')
                n1 = 1;
            else
                n1 = 1, n2 = 1;
        }

        if (q2) {
            if (str[i] == '0')
                n1 = 1;
            else
                n1 = 1, n2 = 1;
        }

        q0 = n0;
        q1 = n1;
        q2 = n2;
    }

    if (q2)
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

## Result:

Thus, the C program to simulate an NFA that accepts binary strings starting with `0` and ending with `1` was written and executed successfully.